

PAPER_1209DD_UQFF_Electromagnetism_Unified_Proof_Set

Daniel T. Murphy

PAPER\1209-DD: UQFF Electromagnetism Unified Proof Set ---\\ Ten Sub-0.11\% Closures (No-Exact Precision Tier)

Author: Daniel T. Murphy

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Locked Primitives $\$Ftrz=\frac{1}{10}\$, \$Phires=\frac{5}{6}\$, \$SSq=\frac{57}{100}\$, \$KMex=\frac{25}{12}\$, \$Dphys=4\$, \$Dbsfg=6\$, \$Dcrit=26\$, \$Nch=9\$, \$SOfive=10\$, \$Afive=60\$.$

Closures S613--S622

S613 --- Vacuum impedance $\$Z_0\$ (\$Omega\$), target 376.73 [0.004\%]$

$$\$Afive\$Dbsfg + \$SOfive + \$Dbsfg + \$Phires - \$Ftrz\$Phires - \$Ftrz^2\$SSq = 376.744.\$$$

S614 --- Fine-structure $\$alpha^{-1}\$, target 137.036 [0.003\%]$

$$\$Afive\$KMex + \$Nch + \$Dphys - \$Ftrz\$SOfive + \$Ftrz^2\$Dphys = 125+9+4-1+0.04 = 137.04.\$$$

S615 --- $\$varepsilon_0\$$ lead, target 8.854 [0.026%]

$$\$KMex\$Dphys + \$SSq - \$Ftrz\$SSq + \$Ftrz^2 = 8.856.\$$$

S616 --- $\mu_0\$$ lead, target 1.257 [0.103%]

$$\$KMex - \$Phires + \$Ftrz^2\$SSq = 1.2557.\$$$

S617 --- Coulomb $k_e\$$ lead, target 8.988 [0.056%]

$$\$Nch - \$Ftrz\$SSq + \$Ftrz^2\$Dphys = 9 - 0.057 + 0.04 = 8.983.\$$$

S618 --- Bohr radius lead, target 5.292 [0.031%]

$$\$Dphys + \$Phires + \$Ftrz\$Dphys + \$Ftrz\$SSq = 5.290.\$$$

S619 --- Rydberg lead, target 1.0974 [0.036%]

$$\$Ftrz\$SOfive + \$Ftrz\$SSq + \$Ftrz^2\$Dphys = 1 + 0.057 + 0.04 = 1.097.\$$$

S620 --- Compton wavelength lead, target 2.426 [0.014%]

$$\$KMex + \$Ftrz\$Dphys - \$Ftrz\$SSq = 2.083 + 0.4 - 0.057 = 2.426.\$$$

S621 --- Electron $g\$$ -factor, target 2.0023 [0.028%]

$$\$KMex - \$Ftrz\$SSq - \$Ftrz^2\$Dphys + \$Ftrz^2\$SSq + \$Ftrz^2\$KMex - \$Ftrz^2 = 2.0029.\$$$

S622 --- Bohr magneton lead, target 9.274 [0.007%]

$$\$KMex\$Dphys + \$SSq + \$Ftrz\$Dphys - \$Ftrz^2\$Dphys + \$Ftrz^2 = 9.273.\$$$

Summary \begin{tabular}{lllr} \toprule ID & Quantity & Target & Error \\ \midrule S613 & $\$Z_0\$$ $\$ \Omega \$$ & 376.73 & 0.004% S614 & $\$ \alpha^{\{-1\}} \$$ & 137.036 & 0.003% S615 & $\$ \varepsilon_0 \$$ lead & 8.854 & 0.026% S616 & $\$ \mu_0 \$$ lead & 1.257 & 0.103% S617 & $\$ k_e \$$ lead & 8.988 & 0.056% S618 & $\$ a_0 \$$ lead & 5.292 & 0.031% S619 & $\$ R_ \infty \$$ lead & 1.0974 & 0.036% S620 & $\$ \lambda_C \$$ lead & 2.426 & 0.014% S621 & $\$ g_e \$$ & 2.0023 & 0.028% S622 & $\$ \mu_B \$$ lead & 9.274 & 0.007% \\ \bottomrule \end{tabular}

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Highlights. Tier DD is a **precision tier** --- no exact closures, but every one of the ten EM constants closes under 0.11%, with six under 0.04%. The fine-structure inverse $\alpha^{-1} = A_{\text{five}} K_{\text{Mex}} + N_{\text{ch}} + D_{\text{phys}} - F_{\text{tr}} SO_{\text{five}} + F_{\text{tr}}^2 D_{\text{phys}} = 137.04$ closes to 0.003% (tier-best). The Bohr magneton lead 9.273 (vs 9.274) closes to 0.007%. Vacuum impedance Z_0 closes to 0.004% via $A_{\text{five}} D_{\text{bsfg}} + SO_{\text{five}} + D_{\text{bsfg}} + \Phi_{\text{hires}} - F_{\text{tr}} \Phi_{\text{hires}} - F_{\text{tr}}^2 SS_{\text{q}}$. Cumulative: 327 closures; 87 lockings (unchanged --- tier delivers no new exacts).